Information Theory Mn908 and MICAS
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Model Answers to Problem Set 1

September 23, 2026

Problem 1 Conditional Mutual Information and Markov Chains

Let X, Y, Z be discrete random variables.

1. Show that
$$I(X;Y \mid Z) = 0$$
if and only if $X - Z - Y$ form a Markov chain, i.e., if and only if
$$P_{XY|Z}(x, y \mid z) = P_{X|Z}(x \mid z)P_{Y|Z}(y \mid z)$$
for every z with $P_Z(z) > 0$.

2. Construct an example of three binary random variables X, Y, Z for which
$$I(X;Y) = 0 \qquad \text{but} \qquad I(X;Y \mid Z) > 0.$$
Compute both quantities explicitly and explain why $X - Z - Y$ does not form a Markov chain.

Model answer. 1. Recall that
$$I(X;Y \mid Z) = \sum_z P_Z(z) I(X;Y \mid Z = z).$$
For every z with $P_Z(z) > 0$,
$$I(X;Y \mid Z = z) = D\big(P_{XY|Z}(\cdot,\cdot \mid z) \big\| P_{X|Z}(\cdot \mid z)P_{Y|Z}(\cdot \mid z)\big).$$
Relative entropy is nonnegative and equals zero if and only if its two arguments are identical. Hence $I(X;Y \mid Z) = 0$ if and only if, for every z with $P_Z(z) > 0$,
$$P_{XY|Z}(x, y \mid z) = P_{X|Z}(x \mid z)P_{Y|Z}(y \mid z).$$
This is exactly conditional independence of X and Y given Z, or equivalently the Markov chain $X - Z - Y$.

2. Let X and Y be independent fair binary random variables and let
$$Z = X \oplus Y,$$
where $\oplus$ denotes addition modulo 2. Since X and Y are independent,
$$I(X;Y) = 0.$$
Given Y and Z, however, X is determined by $X = Y \oplus Z$, while X is still fair when only Z is known. Thus
$$I(X;Y \mid Z) = H(X \mid Z) - H(X \mid Y, Z) = 1 - 0 = 1 \text{ bit}.$$
Therefore X and Y are not conditionally independent given Z, and $X - Z - Y$ is not a Markov chain.

Problem 2 Mixing Increases Entropy

1. Let
$$p = (p_1, \ldots, p_j, \ldots, p_k, \ldots, p_m)$$
be a probability distribution, and fix $\alpha \in [0, 1]$. Define q by leaving all entries except j and k unchanged and setting
$$q_j = \alpha p_j + (1-\alpha)p_k, \qquad q_k = (1-\alpha)p_j + \alpha p_k.$$
Show that $H(q) \geq H(p)$. When $\alpha = \frac{1}{2}$, interpret the operation in words.

2. Let $q = (q_1, \ldots, q_m)$ be a probability distribution for which the endpoint masses $q_1 = a$ and $q_m = b$ are fixed, where $a, b \geq 0$ and $a + b \leq 1$. Find the smallest and largest possible values of $H(q)$. Specify probability distributions that attain the two bounds.

Model answer. 1. Let $\bar{p}$ be obtained from p by interchanging p_j and p_k. Entropy is invariant under permutations, so $H(\bar{p}) = H(p)$. Moreover,
$$q = \alpha p + (1-\alpha)\bar{p}.$$
By concavity of entropy,
$$H(q) \geq \alpha H(p) + (1-\alpha)H(\bar{p}) = H(p).$$
For $\alpha = \frac{1}{2}$, the two masses are replaced by their average.

2. Put $r = 1 - a - b$. The minimum is obtained by concentrating all remaining mass r in one of the $m - 2$ free positions:
$$H_{\min} = H(a, r, 0, \ldots, 0, b) = -a \log a - b \log b - r \log r.$$
The maximum is obtained by spreading r uniformly over the $m - 2$ free positions:
$$q^* = \left(a, \frac{r}{m-2}, \ldots, \frac{r}{m-2}, b\right),$$
and therefore
$$H_{\max} = -a \log a - b \log b - r \log\left(\frac{r}{m-2}\right).$$
Here $0 \log 0$ is interpreted as zero.

Problem 3 Entropy of Functions of a Random Variable

Let X be a discrete random variable taking values in a finite alphabet $\mathcal{X}$, and let $U = f(X)$ for a deterministic function f. Justify the following steps:
$$H(X, U) \overset{(a)}{=} H(X) + H(U \mid X) \overset{(b)}{=} H(X),$$
$$H(X, U) \overset{(c)}{=} H(U) + H(X \mid U) \overset{(d)}{\geq} H(U).$$
Conclude that $H(f(X)) \leq H(X)$.

- Explain briefly why this inequality is intuitively reasonable.

- Characterize precisely when equality $H(f(X)) = H(X)$ holds. Your statement should take into account that some symbols in $\mathcal{X}$ may have probability zero.

Model answer. Steps (a) and (c) are the chain rule for entropy. Since $U = f(X)$ is determined by X,

$$H(U \mid X) = 0,$$

which proves (b). Conditional entropy is nonnegative, so $H(X \mid U) \geq 0$, which proves (d). Hence

$$H(f(X)) \leq H(X).$$

A deterministic function can discard information but cannot create new randomness.

Equality holds exactly when $H(X \mid f(X)) = 0$, i.e., when X can be recovered uniquely from $f(X)$ with probability one. Equivalently, f must be one-to-one on the support $\{x \in \mathcal{X} : P_X(x) > 0\}$.

Problem 4 Mutual Information and Conditional Mutual Information

Consider a sequence of n (dependent) binary random variables $X_1, \ldots, X_n$. The joint law $P_{X_1,\ldots,X_n}$ is described as follows. For every sequence $x_1, \ldots, x_n$ that has an *even* number of 1's,

$$P_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = 2^{-(n-1)}.$$

For every sequence $x_1, \ldots, x_n$ that has an *odd* number of 1's,

$$P_{X_1,\ldots,X_n}(x_1, \ldots, x_n) = 0.$$

1. Compute $I(X_1; X_2)$ when $n = 10$ and when $n = 2$.
2. Compute $I(X_1; X_2 \mid X_3)$ for $n \geq 3$. How does it depend on n?
3. Compute $I(X_1, \ldots, X_{n-1}; X_n)$ and $I(X_1; X_n \mid X_2, \ldots, X_{n-1})$.

Model answer. 1. For $n \geq 3$, every pair (X_1, X_2) is uniform on $\{0,1\}^2$: for each fixed pair there are 2^{n-3} completions with even total parity. Hence X_1 and X_2 are independent and

$$I(X_1; X_2) = 0.$$

In particular, this gives $I(X_1; X_2) = 0$ for $n = 10$. For $n = 2$, only $(0,0)$ and $(1,1)$ are possible, each with probability $1/2$. Thus $X_1 = X_2$ and

$$I(X_1; X_2) = H(X_1) = 1 \text{ bit}.$$

2. If $n \geq 4$, every triple (X_1, X_2, X_3) is uniform on $\{0,1\}^3$, since each fixed triple has 2^{n-4} valid completions. Thus X_1, X_2, X_3 are mutually independent and

$$I(X_1; X_2 \mid X_3) = 0.$$

For $n = 3$, the parity constraint gives $X_1 \oplus X_2 \oplus X_3 = 0$. Given X_3, knowing X_1 determines X_2, while X_1 remains fair. Hence

$$I(X_1; X_2 \mid X_3) = 1 \text{ bit}.$$

Therefore the answer is 1 bit for $n = 3$ and 0 for $n \geq 4$.

3. The last bit is completely determined by the first $n-1$ bits through the even-parity constraint, while X_n itself is fair. Therefore

$$I(X_1, \ldots, X_{n-1}; X_n) = H(X_n) = 1 \text{ bit}.$$

Similarly, after conditioning on $X_2, \ldots, X_{n-1}$, the parity constraint gives a one-to-one relation between X_1 and X_n. Since X_n remains fair given the middle variables,

$$I(X_1; X_n \mid X_2, \ldots, X_{n-1}) = 1 \text{ bit}.$$